

# Ravani Roshan

AI Engineer · Full-Stack Developer · OSS Builder

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## PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate specialising in agentic AI, LLM systems, and production-grade full-stack development. Author of Niki (hermetic multi-agent coding system) and backstop (in-process AI SDK guardrails). Hands-on experience designing end-to-end systems across reasoning agents, RAG pipelines, real-time APIs, and Windows automation. Strong proficiency in Rust, Python, and JavaScript/TypeScript, with emphasis on correctness, reliability, and interpretability.

## EDUCATION

### Silver Oak University

*B.Tech, Computer Science Engineering*

Ahmedabad, Gujarat, India

*CGPA: 8.0*

## EXPERIENCE

### Google Developers Group

*Core Developer*

Oct 2023 – Dec 2025

*Ahmedabad, Gujarat, India*

- Led engineering sessions on AI tooling and agentic system design for a community of 200+ developers; mentored peers on LLM integration and production deployment.
- Contributed to multi-domain software projects spanning web infrastructure and applied ML with focus on correctness, maintainability, and system-level thinking.
- Maintained active engagement with emerging frameworks including LangChain, LangGraph, and Claude API tooling.

## PROJECTS

### Niki – Hermetic Multi-Agent Coding System | *Rust, Docker, CLI, LLM*

Jul 2026 – Present

- Built Niki, a hermetic multi-agent coding system in Rust. AI agents independently plan, code, test and review in Docker sandboxes, then deliver reviewable git branches. Fire-and-forget, BYOK, open source.

### Phantom – Background Computer-Use Agent | *Rust, PyO3, gRPC, Windows API*

Jul 2026 – Present

- Built a background computer-use agent in Rust that drives Windows desktop actions through screenshots, mouse/keyboard, and scroll—without stealing cursor focus.
- Rust core with Python LLM service, gRPC transport, and accessibility-tree integration.

### backstop – In-Process AI SDK Guardrails | *Python, OpenAI SDK, Anthropic SDK, httpx*

Jul 2026 – Present

- Built SDK-native safety layer enforcing token budgets, circuit-breaking, retry policies for OpenAI/Anthropic clients via transport-level interception—no proxy or network hop.
- Wedge multi-agent diff CLI for parallel agent cost isolation without proxy hops.

### LegacyLens – AI Archaeologist for Legacy Codebases | *Gemini 2.0 Pro, Python, Graph Analysis*

Dec 2025

- Transforms 1M+ lines of legacy source code into dynamic dependency graphs using Gemini 2.0 Pro to trace control/data flow and identify structural fragility.

## TECHNICAL SKILLS

**Programming & Systems:** Rust (correctness-driven systems), Python (ML pipelines, agent orchestration, FastAPI), JS/TS (React, Next.js, Node.js), C++

**AI/ML Systems:** LLMs (system design, integration, evaluation), Agents (LangGraph, custom orchestration, tool use), RAG (full stack), Vision-Language models

**Backend & Infrastructure:** FastAPI, PostgreSQL, MySQL, Docker, AWS, GCP, Vercel, Railway, Nginx, Supabase

**Product & Frontend:** React/Next.js dashboards, real-time WebSocket systems, product-oriented engineering

## CERTIFICATIONS

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**AI Systems & Agents:** Claude Code in Action (Anthropic, Mar 2026), AI Fluency Framework (Anthropic, Mar 2026), Transformer Models & BERT (Google, Mar 2026), Responsible AI: Privacy & Safety (Google, Jan 2026), MLOps for GenAI (Google, Dec 2025), AI Agents in LangGraph (DeepLearning.AI), Finetuning LLMs (DeepLearning.AI), LangChain for LLM App Dev (DeepLearning.AI), Prompt Engineering for Vision Models (DeepLearning.AI, Dec 2024)

**ML/Data Science:** ML with Python (freeCodeCamp, Dec 2024), Computer Vision (Kaggle, Nov 2024), Statistics 101 (Cognitive Class, Dec 2024)

**Software Engineering:** Front End Dev Libraries (freeCodeCamp, Feb 2025), Foundational C# (freeCodeCamp, Nov 2024), Forage Job Simulations (EA, Walmart, Goldman Sachs, 2024)